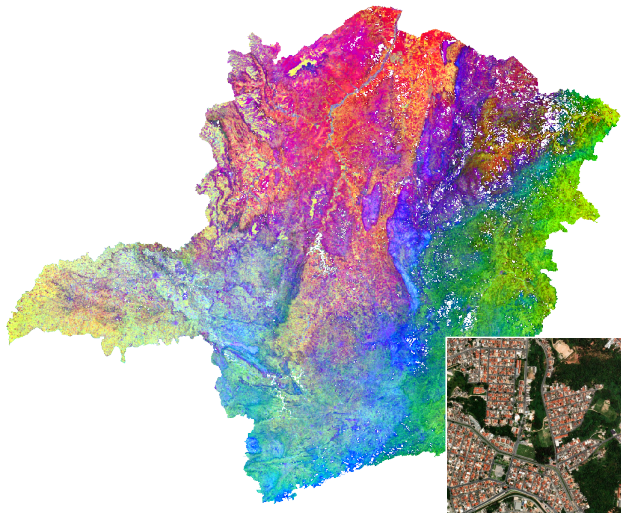


Estimating municipal GDP from satellite embeddings, text and night lights

Combining UrbanCLIP-style captions with
night-light steering of Google satellite
embeddings, tested on the 853 municipi-
palities of Minas Gerais, Brazil

Hemant

Term project • mid-term review • Octo-
ber 2026

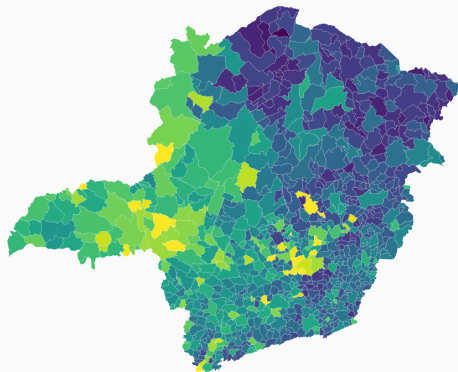


one 1 km cell, as the captioning model sees it

Map: first three principal components of the 64 GSED embedding bands, 1 km cells

Official GDP is patchy below the national level — satellites see everywhere

- Sub-national GDP is published late, estimated from partial data, or missing in many countries
- Satellites measure every place, every year, with the same sensor: an independent signal of economic activity
- **Goal:** estimate municipal GDP per capita from free satellite data, and measure honestly how well it works



R\$ 10k 20k 50k 100k
GDP per capita, 2022 (log scale)

Official 2022 GDP per capita (IBGE), 853 municipalities

We combine two recent ideas that have never been tested together

UrbanCLIP (Yan et al., WWW 2024)

A language model writes a **caption for each satellite tile**; image and text are aligned.

GDP gain only +0.5% R^2 ; its random split lets neighbouring tiles leak into the test.

Satellite embeddings (Yue et al., 2026)

Google's **64-number embeddings, steered by night lights** towards economic meaning.

Beats night lights for GDP levels, but only for countries; no text, no POI/roads.

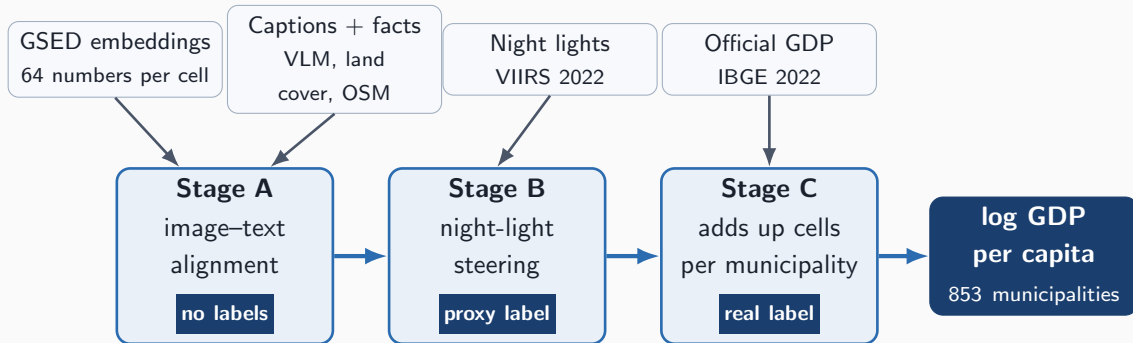
captions

embeddings + steering

This project

Both ideas in one pipeline
Municipal level (853 units)
Honest spatial test + ablations: which ingredient helps?

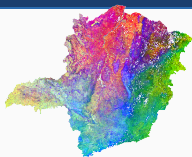
The pipeline: three learning stages, from no labels to official GDP



Captions are used only in training; at prediction time each cell needs only its embedding.

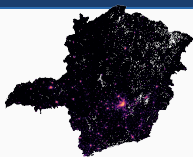
Stages A and B can be switched off to measure what each one adds (variants V3–V9).

We built a 1 km dataset covering all 853 municipalities of Minas Gerais



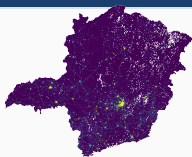
Satellite embeddings

Google GSED, 10 m $\rightarrow$ 1 km



Night lights

VIIRS 2022



Built-up land

ESA WorldCover



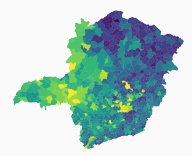
Roads and POIs

OpenStreetMap



Building footprints

Microsoft + OSM



GDP and population

IBGE 2022 (target)

853

municipalities

575k

1 km cells

10,000

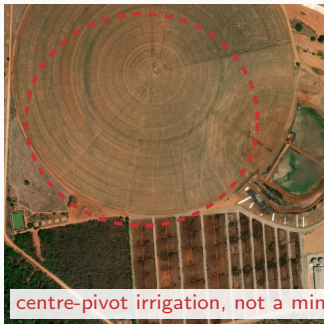
captioned tiles

10M

building footprints

All layers are free and share one 1 km grid. The GDP target is official statistics, not derived from night lights.

A vision-language model describes each tile; data checks remove false claims



Esri World Imagery, ~1 m per pixel

10,000 captions

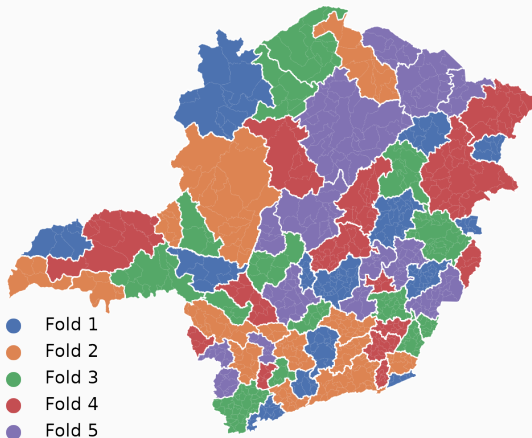
84% of sentences kept

blind hand audit in progress

Qwen2.5-VL-7B caption after data checks

- ~~The satellite image shows a large circular area with concentric rings, likely a mine pit, surrounded by a grid of smaller rectangular plots of farmland.~~ [unsupported industry claim]
- Roads crisscross the area, connecting the plots and the perimeter.
- Vegetation is sparse, mostly along the edges and in some patches within the farmland.
- A small body of water, possibly a pond, is visible near the farmland.
- No buildings or industrial structures are evident, suggesting minimal built-up area.
- ~~The overall scene appears to be an agricultural landscape with a focus on farming and possibly mining operations.~~ [filler]

Every municipality is tested in a region the model never saw

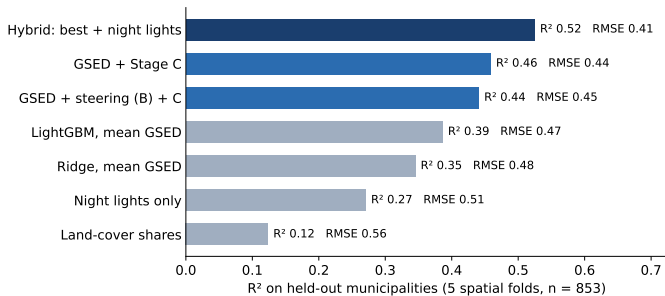


- 70 official regions of neighbouring municipalities → 5 balanced folds
- Each round: 3 folds train, 1 validates, 1 is tested; all stages are retrained
- Nearby places look alike, so holding out whole regions prevents the leakage in UrbanCLIP's random split

Differences between models are tested with a block bootstrap over regions (95% intervals).

First results: the hybrid model explains about half of GDP variation

■ our pipeline ■ baselines



RQ1 steering

no clear effect ($\Delta RMSE +0.007$,
95% CI $[-0.028, +0.041]$)

RQ4 hybrid

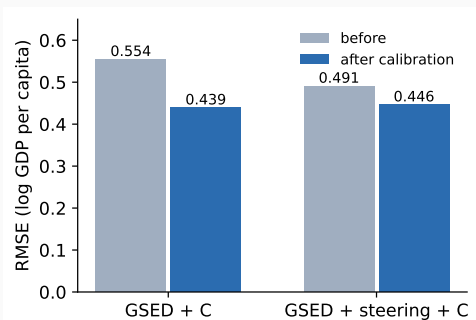
V10 better ($\Delta RMSE -0.028$, 95% CI
 $[-0.050, -0.008]$)

RQ2, RQ3 text, POI/roads

text variants V5–V9 finishing now

Night lights alone: R^2 0.27. Hybrid = our best model blended with the night-light benchmark (weight chosen on validation).

Lessons so far: calibration mattered more than night-light steering



Same models, before and after a validation-fitted correction

1. **Predictions were over-spread.** A straight-line correction fitted on validation municipalities cut error by about 20%.
2. **Honest testing changes conclusions.** Steering looked helpful before the fix; after it, the gain vanished.
3. **Captions need checking.** The model invents mines, roads and farmland on featureless tiles; data checks remove most of it.

About 80% done: text experiments are finishing, full results this week

Data and grid



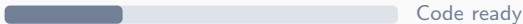
Captions and cleaning



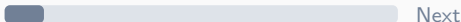
Models and experiments



Error analysis and maps



Report and slides



Research questions

RQ1 steering answered

RQ2 text running

RQ3 POI and roads running

RQ4 hybrid answered

RQ5 where it fails next

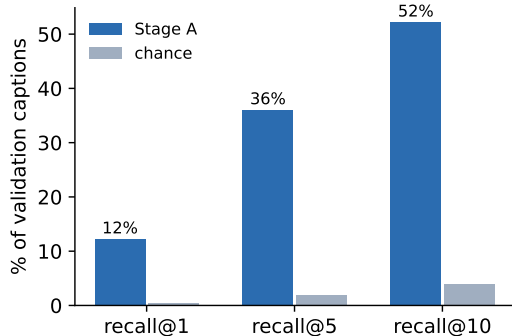
Next

text results → caption audit → error analysis and 1 km maps → report

Backup: data sources

Layer	Source	Use
Satellite embeddings	Google Satellite Embedding V1 (AlphaEarth), 2022	image input, averaged 10 m → 1 km
Night lights	VIIRS DNB annual V2.2, 2022	Stage B proxy label, night-light benchmark
Land cover	ESA WorldCover v200, 2021	masking, fact sentences, caption checks
Roads, POIs	OpenStreetMap (Geofabrik, 2023-01-01)	features, fact sentences, caption checks
Buildings	Microsoft ML footprints + OSM	caption checks
Caption imagery	Esri World Imagery (ArcGIS Location Platform)	input to the vision–language model
GDP, population	IBGE municipal GDP 2022, Census 2022	target (log GDP per capita)

Backup: Stage A aligns embeddings and captions far above chance



For a validation tile, how often is its own caption among the top k of 256 candidates?

Part of the alignment comes from the fact sentences; variants V8 (captions only) and V9 (facts only) separate the two.

Backup: what the captions look like



Forest

A predominantly forested area with patches of cleared land.



Farmland

A large, circular area of brown farmland, likely recently plowed, dominating the center.



Village edge

A sparsely built area with scattered houses and small buildings, likely residential.



Town centre

A densely built-up area with numerous houses featuring red-tiled roofs, indicating a residential neighborhood.

First sentence of each caption that passed the cleaning checks.